

Morgan Patel

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EDUCATION

Ridgeway University | Ph.D. candidate in Statistics | Metro City, USA | Sep 2021 - Expected May 2026

Ridgeway University | B.S. in Mathematics | Metro City, USA | Sep 2016 - May 2020

EXPERIENCE

Northpeak Capital | Quantitative Research Intern | Metro City, USA | Jun 2025 - Aug 2025

- Built a short-horizon order-book imbalance signal for liquid index futures that added 0.4 Sharpe to the desk's book over 18 months of out-of-sample backtest after costs.
- Cut the signal's daily turnover from 34% to 21% with a cost-aware position smoother, keeping 90% of gross returns and lowering estimated slippage by a third.
- Validated the signal on 6 years of tick data, using a Sharpe-ratio test for heteroskedasticity to confirm there was no look-ahead leakage.
- Joining 120 microstructure features point-in-time across six venues, deduplicating late prints and versioning each schema, built a feature store the team reused in two later projects.
- Tuned the signal's parameters on the full 6-year sample, then reported the Sharpe of a walk-forward test over the same period as out-of-sample.
- Documented the backtest assumptions, transaction-cost model and known failure regimes in the desk's research wiki for future interns.

Ridgeway University | Graduate Research Assistant, Statistical Learning Lab | Metro City, USA | Feb 2021 - Aug 2021

- Owned the lab's simulation pipeline on the shared cluster and the reproducibility checks other students relied on.
- Derived a variance bound for a sparse regression estimator that tightens the previous bound by a log factor; it is now Section 3 of a paper under review at JASA.
- Taught weekly recitations for 60 students in graduate probability, writing 12 problem sets and earning a 4.8/5 teaching rating.
- Released an open-source R package for high-dimensional covariance estimation while maintaining the lab's shared cluster, organizing the weekly reading group and grading for two courses, which was downloaded 3,000 times in its first year.

PROJECTS

Volatility Forecasting Study | Independent Research | Python, PyTorch | Jan 2024 - Present

- Beat a HAR-RV baseline's out-of-sample QLIKE loss by 7% on 30 equity indices with a temporal convolutional model trained on realized-volatility features.
- Cut forecast error from 0.20 to 0.15, a 33% improvement, by adding realized-volatility features and an asymmetric loss.
- Improved the model's directional hit rate by 6% on 30 equity indices, from 52% to 58%, with a temporal convolutional model.

Kaggle Market Prediction Competition | Team of 3 | Python | Mar 2023 - Jun 2023

- Placed 41st of 2,900 teams with a gradient-boosting ensemble on 300 engineered features, finishing in the top 2% of the private leaderboard.
- Cut validation leakage by switching to time-grouped folds, which closed a 0.02 gap between local validation and leaderboard scores.
- Wrote the team's feature-selection script using permutation importance, trimming 900 candidate features to 300 without losing validation score.

SKILLS

Programming: Python, R, SQL, C++, Kafka

Methods: time-series econometrics, Bayesian inference, gradient boosting, PyTorch